

Lecture 5

Semaphores and Mutexes

The POSIX thread library contains functions for working with semaphores and mutexes. There is much more to say than what is mentioned here. A good place to find more information is <https://linux.die.net>.

The functions should all be compiled and linked with `-pthread`.

What is a semaphore in LINUX?

(Library: `#include <semaphore.h>`)

A semaphore is fundamentally an integer whose value is never allowed to fall below 0. There are two operations on a semaphore: wait and post. The post operation increments the semaphore by 1, and the wait operation does the following: If the semaphore has a value > 0 , the semaphore is decremented by 1. If the semaphore has value 0, the caller will be blocked (busy-waiting or more likely on a queue) until the semaphore has a value larger than 0, and then it is decremented by 1. We declare a semaphore as:

`sem_t sem;`

where `sem_t` is a typedef defined in a header file as (apparently) a kind of unsigned char.

An example of this might be that we have a set of N interchangeable resources. We start with semaphore $S = N$. We use a resource, so there are now $N-1$ available (wait), and we return it when we are done (post). If the semaphore has value 0, there are no resources available, and we have to wait (until someone does a post).

Semaphores are thus used to coordinate concurrent processes.

This is what some people call a "counted semaphore". There is a similar notion called a "binary semaphore" which is limited to the values 0 and 1.

A semaphore may be named or unnamed. These notes assume we are using named semaphores.

Semaphore Functions in C

1. `int sem_init(sem_t * sem, int pshared, unsigned int value);`

Purpose:

This initializes the semaphore `*sem`. The initial value of the semaphore will be `value`. If *pshared* is 0, the semaphore is shared among all threads of a process (and hence need to be visible to all of them such as a global variable).

If *pshared* is not zero, the semaphore is shared but should be in shared memory.

Notes:

- On success, the return value is 0, and on failure, the return value is -1.
- An attempt to initialize a semaphore that has already been initialized results in undefined behavior.

2. `int sem_wait(sem_t * sem);`

Purpose: This implements the wait function described above on the semaphore `*sem`.

Notes:

- Here `sem_t` is a typedef defined in the header file as (apparently) some variety of integer.

- On success, the return value is 0, and on failure, the return value is -1 (and the value of the semaphore is unchanged).
 - There are related functions `sem_trywait()` and `sem_timedwait()`.
3. **int sem_post(sem_t * sem);**
Purpose: This implements the post function described above on the semaphore *sem.
Note: On success, the return value is 0, and on failure, the return value is -1 (and the value of the semaphore is unchanged).
4. **int sem_destroy(sem_t * sem);**
Prototype: `int sem_destroy(sem_t * sem);`
Purpose: This destroys the semaphore *sem, so *sem becomes uninitialized.
Notes:
- On success, the return value is 0, and on failure, the return value is -1.
 - Destroying a semaphore on which other processes or threads are waiting (using `sem_wait()`) or destroying an uninitialized semaphore will produce undefined results.

What is a mutex in LINUX?

(Library: `#include <pthread.h>`)

A mutex (named for "mutual exclusion") is a binary semaphore with an ownership restriction: it can be unlocked (the post operation) only by whoever locked it (the wait operation). Thus a mutex offers a somewhat stronger protection than an ordinary semaphore.

We declare a mutex as:

`pthread_mutex_t mutex;`

mutex Functions in C

- 1) **int pthread_mutex_init(pthread_mutex_t * restrict mutex, const pthread_mutexattr_t * restrict attr);**
Purpose: This initializes *mutex with the attributes specified by attr. If attr is NULL, a default set of attributes is used. The initial state of *mutex will be "initialized and unlocked".
Notes:
- If we attempt to initialize a mutex already initialized, the result is undefined.
 - On success, the return value is 0, and on failure, the return value is a nonzero value indicating the type of error.
 - In the prototype, the keyword restrict (part of the C99 standard) means that this pointer will be the only pointer to the object.
- 2) **int pthread_mutex_destroy(pthread_mutex_t * restrict mutex);**
Purpose: This destroys the mutex object *mutex, so *mutex becomes uninitialized.
Notes:
- It is safe to destroy an unlocked mutex but not a locked mutex.
 - The object *mutex could be reused, i.e., reinitialized.
 - On success, the return value is 0, and on failure, the return value is a nonzero value indicating the type of error.
- 3) **int pthread_mutex_lock(pthread_mutex_t * mutex);**

Purpose: This locks *mutex. If necessary, the caller is blocked until *mutex is unlocked (by someone else) and then *mutex is locked. When the function call ends, *mutex will be in a locked state.

Notes:

- Suppose we try to relock a locked mutex. Depending on the attributes of the mutex, we may have an error, or a count may be kept of how many times the caller has locked the same mutex (and thus will have to unlock it the same number of times).
- On success, the return value is 0, and on failure, the return value is a nonzero value indicating the type of error.

4) int pthread_mutex_unlock(pthread_mutex_t * mutex);

Purpose: This unlocks *mutex.

Notes:

- Suppose we try to unlock an unlocked mutex. Depending on the attributes of the mutex, we may have an error.
- On success, the return value is 0, and on failure, the return value is a nonzero value indicating the type of error.